

Technical Report — Fact Attack: Multilingual NLI over Swiss Voting Booklets

- **Track:** Track 2A — OST: Multilingual Natural Language Inference over Swiss Official Voting Booklets
- **Event:** Hack Apertus Online (October 1–16, 2026)
- **Team:** Team Fact Attack 2026 — Josip Jukic, Felipe Wüthrich
- **Repository:** [GitHub](#) · **Experiment log:** [docs/experiments.md](#)
- **Demo:** Streamlit app (make web: claim check with highlighted evidence pages, benchmark dashboard) and the entailment CLI (python -m src --input cases.jsonl --output predictions.jsonl, the Docker image's entrypoint, see 2.5)

1. Summary

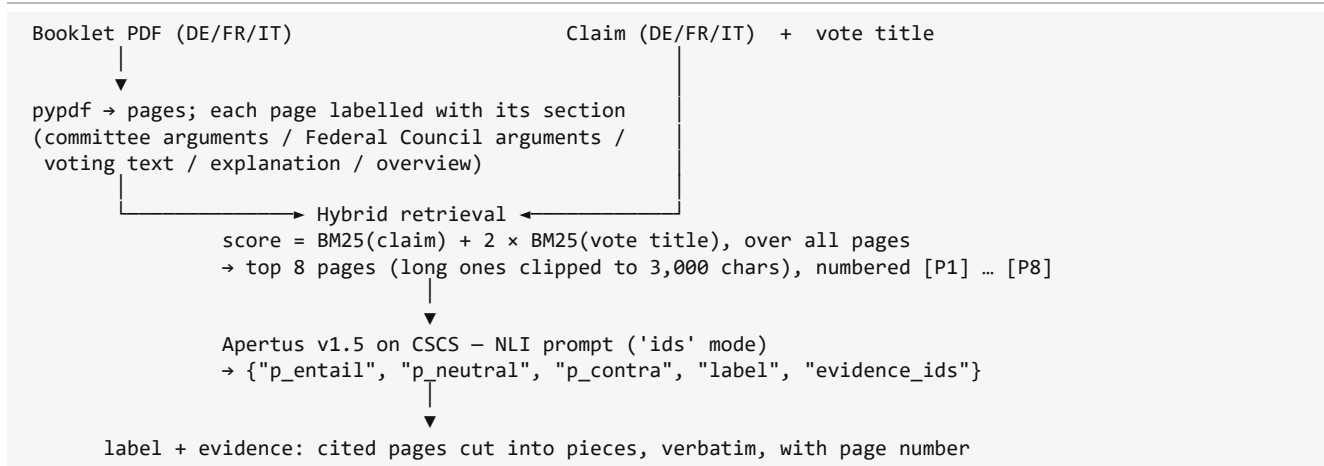
Swiss voters receive an official booklet for every federal vote, and campaigns make claims about it that are hard to check. We built a multilingual claim-verification system on **Apertus v1.5**: given a booklet (German, French or Italian) and a claim (any of the three languages), it classifies the claim as **Entailment (0)**, **Neutral (1)** or **Contradiction (2)** and returns the booklet pages that justify the decision, together with token usage and inference time.

On a test split of **5 voting dates never used during development** (401 pairs of the official dataset v1.1, scored with the organisers' `evaluate.py`), the system reaches **macro-F1 0.930** and **evidence Hit@5 0.727** on the advanced task (booklet PDF) and **macro-F1 0.973** on the beginner task (reference text), at 3,877 / 2,532 input tokens, ~57 output tokens and ~5 ms of processing time per case outside the LLM calls.

The official score also weighs evidence, processing time and tokens (relative to the best team), so after the organisers' Q&A we re-tuned for it (section 2.6): 8 instead of 12 pages save 31 % of the input tokens for -0.015 F1 on test (-0.003 on dev), and evidence made of page pieces raised Hit@5 from 0.34 to 0.73 at no cost. With 12 pages the advanced task reached 0.945 on test and 0.946 over all 1,495 pairs.

The central finding for the challenge's research question: **sending a dozen well-chosen pages beats sending the whole booklet** — 0.946 vs. 0.730 macro-F1 on the same 150 test pairs, with ~10× fewer input tokens.

2. Architecture



2.1 Document processing

`pdf_parser.py` extracts the text of every page with `pypdf` and detects section headings in all three languages ("Argumente Initiativkomitee", "Arguments du Conseil fédéral et du Parlement", "Il testo in votazione", ...). A heading applies to its page and the following pages until the next heading, and table-of-contents entries (heading followed by a page number) are ignored. Section labels matter because booklets deliberately contain opposing viewpoints: the model has to know whether a page states the committee's or the Federal Council's position.

2.2 Context selection: hybrid retrieval

`retriever.py` scores every page of the booklet with normalised BM25 against the claim plus twice the normalised BM25 against the vote title, and keeps the top 8 pages (12 before the Q&A, see 2.6). Pages longer than 3,000

characters are clipped to the window that best matches claim and vote title (plus the page heading).

- The **vote title is written in the booklet's language**, so it anchors the right proposal even when the claim is in another language; lexical matching of the claim alone fails for cross-lingual pairs.
- Our first version filtered hard to one detected proposal. That dropped the per-proposal overview pages at the front of the booklet, where the gold passage sits for ~11 % of pairs, and chose the wrong proposal for another ~10 %. Booklet-wide scoring raised gold-section recall on the dev split from 0.74 to 0.93.
- **Speaker boost**: if the claim names a side ("the initiative committee argues ...", "selon le Conseil fédéral ..."), the two best argument pages of that side are always among the selected pages, replacing the lowest-ranked others. Without it, the committee's own page was missing in 15 of 36 committee-claim errors.

2.3 Apertus as the NLI model

The selected pages are numbered [P1]...[P8]. The system prompt defines the three labels, tells the model to judge only against the passages (not world knowledge), to answer Neutral for topics or details the passages do not address, to check claims attributed to an actor only against that actor's text, and to read "recommends rejecting X" as supported when the Federal Council and Parliament recommend "No". Apertus answers with a small JSON object: three confidences, the label, and the ids of the pages that justify it.

- **Evidence by id** guarantees verbatim booklet text with the correct page number and avoids paraphrased quotes.
- **Short answers**: ~55 output tokens instead of ~150 for a JSON answer with reasoning and quotes, which more than halves inference time (p95 5.5 s → 1.8 s on the dev split) at equal or better F1.
- **Evidence items**: the first two cited pages are returned in halves, the free slots hold the best-matching third of further pages (cited, then retrieved); five verbatim items. Hit@5 counts an item only if it lies inside the gold passage or contains it, which a whole page rarely does (it also holds a heading or the next section).
- For **Neutral**, the evidence list is empty. If the model predicts 0 or 2 without citing a page, the top-ranked page is used.
- **Decision rules**: in ~2 % of answers the label and the stated confidences disagree. The label is right 3× as often, so it is kept (Rule 0) — except a Neutral label next to a confident other relation, which was right in only 1 of 18 cases and follows the confidences (Rule 0b). Otherwise label and confidences agree and are used as given.

2.4 Beginner task

The reference string (up to 25k characters) is split into passages of ≤ 700 characters at sentence boundaries, numbered and sent with the same prompt, so evidence is a specific passage rather than the whole reference.

2.5 CLI and output format

`python -m src --input cases.jsonl --output predictions.jsonl` (the Docker entrypoint) reads the official case format (JSON or JSONL; a case has `booklet.path` + `vote` for the advanced task or `reference.text` for the beginner task) as well as rows of the OST dataset on Hugging Face (`claim`, `reference_string`, `booklet_url`, `vote`; also Parquet or CSV), and writes one record per case:

```
{"id": "case-0042", "label": 2, "label_name": "contradiction",
 "evidence": [{"page": 7, "text": "<first third of page 7>"}, {"page": 7, "text": "<second third>"}, ...],
 "metrics": {"input_tokens": 3877, "output_tokens": 57, "inference_time_ms": 1917}}
```

Evidence is empty for Neutral, and its page is `null` for a reference text. The claim language is taken from the case (or detected), and a failing case is reported on `stderr` and returned as a valid Neutral record instead of aborting the batch; the exit code is 0. `python -m src predict` checks a single claim.

2.6 Evaluation contract and efficiency

The organisers run the image as `--input /data/cases.jsonl --output /output/predictions.jsonl` with `BASE_URL` (a token-counting proxy) and `API_KEY` injected, `/data` read-only. The score is $0.6 \cdot S_A + 0.4 \cdot S_B$, where $S_A = 50$ % macro-F1 + 20 % evidence Hit@5 + 15 % processing time + 15 % tokens and $S_B = 70$ % macro-F1 + 15 % time + 15 % tokens; time and tokens count as best ÷ own over all teams, and processing time is the wall clock minus the time with an Apertus request in flight.

- **Contract**: `BASE_URL` / `API_KEY` take precedence over our older variable names; the parse cache of the 60 dataset booklets is baked into the image read-only and new entries go to `/tmp`; a CI step runs the image exactly this way.

- **Tokens:** 8 pages instead of 12 (-31 % input tokens); a condensed prompt was rejected (5.3).
- **Processing time:** cases run four at a time, so parsing and retrieval of one case overlap with requests of others; uncached booklets are parsed by a background thread; pypdf and rank_bm25 are imported only when needed. The CLI reports its own estimate of the scored time: ~4 ms per case with cached booklets (test split), ~650 ms when 40 cases need 22 uncached booklets.

3. Use of Apertus

- **Model:** `swiss-ai/Apertus-v1.5-70B-thinking` (the model our CSCS key is authorised for; the thinking phase is not triggered by our prompt, answers start directly with the JSON object).
- **Where it runs:** CSCS inference endpoint (<https://api.inference.cscs.ch/v1>), OpenAI-compatible chat completions, temperature 0, retries with exponential backoff.
- **Role:** Apertus is the only model in the pipeline. It performs the NLI classification and selects the evidence pages; retrieval is lexical (BM25) and runs locally. Apertus 8B v1.5 was used only for a comparison run (5.3); no other model was used for development or evaluation.

4. Data and evaluation protocol

- **Dataset:** `OSTswiss/MNLIOverSwissVotingBooklets` — 1,495 human-annotated claim/reference pairs from 60 booklets (2020–2026), balanced labels, ~66 % cross-lingual pairs.
- **Booklets:** downloaded from bk.admin.ch via the `booklet_url` field (`python -m src.hf_dataset`).
- **Split by voting date**, so all three language versions of a booklet stay on the same side:
- **dev** (1,093 pairs) for all design decisions;
- **test** (402 pairs, 5 voting dates: 2021-06-13, 2021-09-26, 2022-09-25, 2024-06-09, 2024-09-22), used only for before/after measurements. The two booklets our first version was tuned on are pinned to dev.
- **Metrics** (mirroring the judging criteria): macro-F1 over the three labels (primary), macro-F1 per claim→booklet language pair, share of cited evidence inside the gold reference section, input/output tokens, mean and p95 inference time. Every run is saved with its configuration and git commit in `results/`.
- **Official scorer:** after the Q&A, cases were generated with the starter repo's `prepare_cases.py` (dataset v1.1) and scored with its `evaluate.py` (macro-F1 per task, evidence Hit@5), on the same test split (401 pairs) and on 300 random dev pairs; predictions and reports are in `results/official/`.

5. Results

5.1 Final configuration — official scorer, test split (401 pairs)

Task	Configuration	Macro-F1	Hit@5	Input / output tokens	Non-LLM time per case
Advanced	12 pages, whole cited pages (before the Q&A)	0.945	0.341	5,598 / 59	–
Advanced	8 pages, cited pages in halves + thirds of further pages	0.930	0.727	3,877 / 56	~5 ms
Beginner	unchanged	0.973	–	2,532 / 56	~3 ms

Before the Q&A (own evaluator, 12 pages, 402 test pairs) the advanced task scored 0.940 (mono-lingual 0.951, cross-lingual 0.934; weakest pair French claims against German booklets) and the beginner task 0.975; per language pair tables are in the [experiment log](#).

5.2 Full document vs. selected context (advanced, same 150 test pairs)

Context	Prompt	Macro-F1	Input tokens	Output tokens	Latency p95
Full booklet	JSON	0.730	59,517	202	43.7 s*
Proposal filter, top 5 pages	JSON	0.795	3,408	157	36.2 s*
Proposal filter, top 10 pages	ids	0.880	6,502	54	4.1 s
Hybrid, top 10 pages	ids	0.926	6,023	55	3.8 s
Final (12 pages, clipping, speaker boost, decision rules)	ids	0.946	5,713	55	2.8 s

* measured while three runs shared the endpoint.

The full booklet is both the most expensive and the least accurate option: booklets reach 250k characters (~70k tokens), and the relevant passage gets lost in the middle. Selecting context is therefore not only an efficiency measure but also improves NLI quality.

5.3 Ablations on the dev split

Change	Effect	Decision
ids answer instead of JSON with reasoning + quotes	beginner 0.973 → 0.980, advanced 0.786 → 0.797, p95 5.5 s → 1.8 s	adopted
Hybrid retrieval instead of proposal filter	advanced 0.797 → 0.862	adopted
Viewpoint and recommendation rules in the prompt	advanced 0.862 → 0.870, beginner 0.980 → 0.993	adopted
Decision-Rule 0: keep the label when it disagrees with the confidences	44 fixed / 13 broken over all saved runs; test 0.920 → 0.928	adopted
Section labels from bare speaker lines and closing recommendation boxes	Federal Council pages labelled as committee 27 → 0; F1-neutral, evidence grounding 0.86 → 0.88	adopted
Clip pages > 3,000 characters to the best-matching window	test 0.925 → 0.928, input tokens 5,725 → 4,696 (p95 12.7k → 7.4k)	adopted
12 instead of 10 pages (after clipping)	all 1,495 pairs 0.912 → 0.931 (51 fixed / 24 broken); +~890 input tokens	adopted
Speaker boost (2.2)	all pairs 0.931 → 0.940 (22 fixed / 8 broken), committee-claim errors 36 → 27, no extra tokens	adopted
Decision-Rule 0b (2.3)	all pairs 0.940 → 0.945 (8 fixed / 1 broken)	adopted
600-character passages instead of pages (n = 450, paired)	0.831 vs. 0.908; better evidence precision, fewer tokens	rejected
Thinking before answering (prefilled reasoning marker)	dev 450: 0.908 → 0.929 (20 fixed / 11 broken); ~1,300 output tokens and ~23 s per claim instead of 59 and ~2 s; a 300-token budget is worse	rejected (efficiency)
Translate cross-lingual claims into the booklet language first	dev 0.908 → 0.919, test 0.925 → 0.923 (14 fixed / 11 broken); +0.6 s	rejected (noise)
Other fixes for speaker attribution: second verification pass, hiding the opposing side, few-shot examples, speaker hint in the prompt, flipping committee contradictions	each broke at least as many answers as it fixed (verification pass: 0 fixed / 24 broken)	rejected
Other answer formats and filters: label-only answer with logprobs, one-sentence reasoning, confidence thresholds, numerical override, meta-classifier	no gain, or worse (label-only: beginner 0.724); the meta-classifier's only useful split became Rule 0b	rejected
Apertus 8B instead of 70B (450 dev pairs)	0.851 vs. 0.908 F1, p95 1.2 s vs. 4.5 s	70B kept (quality first; Apertus latency is not scored)
Evidence: cited pages in pieces (Hit@5 replayed on real citations; 3, 2 vs. whole pages)	test 0.341 → 0.625, dev 300 0.483 → 0.706	adopted
Evidence: free slots for the best third of further pages, 2, 2	replay dev 300 0.763 → 0.810; real runs dev 0.792, test 0.727	adopted
8 instead of 12 pages (dev 300, official scorer)	F1 0.937 → 0.934, Hit@5 0.70 → 0.76, input tokens 5,370 → 3,721; k=7 / 6: F1 0.924 / 0.904; test F1 0.945 → 0.928	adopted (tokens are scored)
Condensed system prompt (PROMPT_SHORT)	dev 300 B 0.980 → 0.850, A 0.937 → 0.781, 6–13 % fewer input tokens	rejected

Full tables: [docs/experiments.md](#). Runs on 150 pairs carry about ±3 F1 points of sampling noise; close decisions were re-run on 450 pairs.

5.4 What we learned about Apertus

- **Selected context beats more context.** With the whole booklet (~60k tokens) Apertus 70B reaches 0.73 macro-F1, with 10 selected pages 0.93. We found no position effect ("lost in the middle"): the full-document runs fail even when the answer is in the first third of the booklet, so the degradation is general distraction.
- **Apertus is over-confident.** Its self-reported confidences are 0 or 1 in 97–99 % of cases, and the probability of the label token is ~1.0 for wrong answers as well as for correct ones (41 wrong vs. 40 correct dev answers). Confidence thresholds cannot filter its errors.
- **Speaker attribution was its main weakness — mostly a retrieval problem.** Claims attributed to the committee failed in 13.4 % of cases (2.2 % for Federal Council claims), and no prompt change helped. The speaker boost and Rule 0b brought them down to 7.4 %, the same rate as claims without a speaker.
- **Thinking helps, at a high price.** Letting the thinking model reason before its answer fixes half of the remaining dev errors (+0.021 F1 on dev 450), but needs ~22× the output tokens and ~11× the latency. A prompt instruction alone does not make it think first; the reasoning marker has to be prefilled. Selective thinking saves little (it is needed on ~3/4 of claims), and a short hard budget is worse.
- **Cross-lingual asymmetry.** Over all 1,495 pairs, French claims against German booklets are the hardest pair (0.880); French claims against Italian booklets reach 0.986. Small samples mislead here: on the 43 test pairs it→de scored 0.80, over all 148 pairs 0.925.
- **8B vs. 70B.** Apertus 8B is ~3.5× faster (p95 1.2 s) but 6 F1 points weaker, mostly on cross-lingual pairs and contradictions.

6. Limitations

- **French claims against German booklets** are the weakest pair (0.880 over all pairs); their errors spread over many voting dates without a pattern, so no language-specific rule was added.
- **Remaining errors** over all 1,495 pairs (81): Entailment → Contradiction 35, Contradiction → Neutral 22, Contradiction → Entailment 11, Entailment → Neutral 8; in nearly all of them the gold page is in the context. Thinking fixes about half of such errors, at ~11× the latency.
- **Evidence:** Hit@5 is 0.73 on test; most misses cite another page with the same fact than the annotated one.
- **Fewer pages cost some F1 on test** (0.945 → 0.930, -0.003 on dev); the trade-off is right under the scoring formula unless the most frugal team needs less than ~1/8 of our tokens.
- **Layout:** section headings are detected from text lines and recommendation boxes; committee pages headed only by a slogan stay unlabelled. Tables and charts are read as plain text. Layout-aware parsing (Docling) was not adopted: the remaining errors are reasoning errors with the gold page in context, and it would add several GB of models to the image.
- **Processing time and tokens** are measured by the organisers' proxy; our own numbers are estimates (client-side token counts, in-flight clock). Booklets outside the 60 dataset booklets must be parsed at run time (~1 s each).
- **Not political advice:** outputs describe the relationship between a claim and the official booklet only.

7. Reproducibility

```
export API_KEY="your_api_key_here"    # LLM_API_KEY works too
make run                             # Docker build + benchmark on the 402-pair test split (~4 min)
make test                            # unit tests, no API calls

cd track_2a
python -m src.hf_dataset              # dataset, booklets, dev/test split
python -m src.benchmark -d data/hf/test.jsonl -t advanced # writes results/<timestamp>_....json
python -m src.benchmark -d data/hf/test.jsonl -t beginner
python -m src.results_summary         # results/summary.json (used by the app)
python -m src.warm-cache              # pre-parse booklets (disk cache keyed by file hash)
```

The official contract (--input/--output, BASE_URL, API_KEY) is shown in the README; the official-scorer runs are in results/official/. Temperature 0, so repeated runs differ by at most a few pairs.

Configuration is read from environment variables: LLM_NAME, BASE_URL / API_KEY (OR LLM_BASE_URL / LLM_API_KEY), NLI_STRATEGY (hybrid), NLI_TOP_K (8), PAGE_MAX_CHARS (3000), PROMPT_MODE (ids), PASSAGE_CHARS (0 = pages), EVIDENCE_SPLIT (2,2), EVIDENCE_FILL (true), NLI_WORKERS (4).

8. Next steps

1. Error analysis of the remaining Entailment → Contradiction cases (35 of 81 errors), where the gold page is in the context.
2. Evidence that follows the gold passage more closely (e.g. pieces that may cross a page break, or the best-matching window per cited page), keeping whole pages as model context.
3. Reasoning at lower cost: full thinking gains +0.02 F1 at $\sim 11\times$ latency; fine-tuning a smaller Apertus on the reasoning traces could keep the gain without the latency.
4. Multilingual dense retrieval (e.g. BGE-M3) and layout-aware parsing (e.g. Docling) — low priority: the remaining errors are mostly reasoning errors with the gold page retrieved.

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